

1a. $s = \tau = 0 \quad \ddot{x} + \alpha x + \beta x^3 = 0$

$$\ddot{x} = -\alpha x - \beta x^3$$

$$F(x) = m\ddot{x} = m(-\alpha x - \beta x^3) = -m\alpha x - m\beta x^3$$

$$F(x) = -\frac{dU}{dx} = -m\alpha x - m\beta x^3$$

$$U(x) = \int_0^x (m\alpha x + m\beta x^3) dx + U_0$$

Take U to be zero where $x=0$ (Force is zero at $x=0$)

$$U(x) = \frac{m\alpha x^2}{2} + \frac{m\beta x^4}{4} \quad \text{gives the undriven and undamped duffing oscillator EOM}$$

1b. $\frac{dU}{dx} = m\alpha(2x) + \frac{m\beta}{4}(4x^3) = m\alpha x + m\beta x^3 = 0$

$$m x (\alpha + \beta x^2) = 0 \quad x=0, x = \sqrt{-\frac{\alpha}{\beta}}, x = -\sqrt{-\frac{\alpha}{\beta}}$$

$$\alpha + \beta x^2 = 0$$

$$x^2 = -\frac{\alpha}{\beta}$$

$$\frac{d^2U}{dx^2} = m\alpha + 3m\beta x^2$$

$U(0) = m\alpha$ If $\alpha > 0$ then $U(0) > 0$ hence $x=0$ is a stable equilibrium point

If $\alpha < 0$ then $U(0) < 0$ hence $x=0$ is an unstable equilibrium point

$$U\left(\pm\sqrt{-\frac{\alpha}{\beta}}\right) = m\alpha + 3m\beta\left(-\frac{\alpha}{\beta}\right) = m\alpha - 3m\alpha = -2m\alpha$$

If $\alpha < 0$ then $U\left(\pm\sqrt{-\frac{\alpha}{\beta}}\right) > 0$ hence $x = \pm\sqrt{-\frac{\alpha}{\beta}}$ are stable equilibrium points

If $\alpha > 0$ then $U\left(\pm\sqrt{-\frac{\alpha}{\beta}}\right) < 0$ hence $x = \pm\sqrt{-\frac{\alpha}{\beta}}$ are unstable equilibrium points

This assumes either α or β is negative and the other positive for $x = \pm\sqrt{-\frac{\alpha}{\beta}}$ to exist.